

Federated Averaging for Privacy-Preserving Transportation Mode Classification Using Mobility Data

Carla Jaffal
VIP mobisense
American University Beirut
carlajaffal@gmail.com

Abstract

Transportation mode detection is an important task in intelligent transportation systems and smart mobility analysis. Traditional machine learning approaches often require centralized data collection, which raises privacy concerns when dealing with mobility data from multiple geographic locations. Federated learning provides a distributed alternative where models are trained collaboratively without sharing raw data.

In this project, we implement a federated learning framework for transportation mode classification using mobility datasets collected from multiple cities. Each city acts as a federated client participating in the training process through the Federated Averaging (Fed-Avg) algorithm. A baseline federated multilayer perceptron model is trained across clients, followed by additional experiments involving class balancing and ensemble learning techniques.

Experimental results demonstrate the feasibility of federated learning for distributed mobility classification while highlighting challenges related to feature representation and dataset heterogeneity.

Federated Learning, Transportation Mode Detection, Mobility Data, Fed-Avg, Machine Learning

I. INTRODUCTION

Transportation mode detection plays a critical role in modern intelligent transportation systems. By analyzing mobility data collected from GPS devices and mobile sensors, machine learning models can infer user transportation modes such as walking, driving, or cycling. These models are useful for applications including traffic analysis, urban planning, and smart mobility services.

Traditional machine learning approaches require collecting all data in a centralized location for training. However, mobility data is often distributed across multiple geographic regions and organizations, raising concerns related to data privacy and data ownership.

Federated Learning (FL) has emerged as a promising paradigm for distributed machine learning. In federated learning, multiple clients collaboratively train a shared global model without exchanging raw data. Instead, each client performs local training on its private dataset and sends model updates to a central server for aggregation.

One of the most widely used federated learning algorithms is Federated Averaging (FedAvg), which aggregates client model parameters through weighted averaging. This approach enables collaborative model training while preserving data privacy.

In this project, we implement a federated learning framework for transportation mode classification using mobility datasets collected from different cities. Each city dataset is treated as an independent client in the federated learning environment. The objective is to evaluate the effectiveness of the Fed-Avg algorithm for distributed transportation mode detection and explore extensions including balanced training and ensemble-based learning approaches.

II. DATASET AND PREPROCESSING

The dataset used in this project consists of mobility records collected from multiple geographic regions. Each region corresponds to a different city dataset. To simulate a federated learning environment, each city dataset is treated as an independent client participating in the federated training process.

The following cities were used as federated clients:

- Jeddah
- KAUST
- KZ
- Mekkah

Each dataset contains GPS trajectory information including spatial and temporal attributes describing user movement.

A. Transportation Mode Selection

The original datasets contained multiple transportation modes. However, not all transportation modes were present in every city dataset. To ensure consistent classification across all clients, only the modes that appeared in all datasets were selected.

The common transportation modes identified were:

- Car
- Walk

The problem was therefore formulated as a binary classification task.

A. Feature Selection

Initial datasets contained a large number of features, many of which contained significant amounts of missing values. To improve data quality and ensure consistent features across all clients, features with more than 60% missing values were removed.

After filtering, the remaining common features used for training were:

- Latitude
- Longitude
- Time seconds

These features capture spatial and temporal information describing user mobility patterns.

B. Data Splitting

Each client dataset was divided into training and testing subsets using an 80/20 split. The training data was used for federated model training while the test data was reserved for evaluating model performance.

The dataset sizes for each client are summarized in Table I.

client	Table samples	Test samples
Jeddah	569,388	142,347
KAUST	1,337,492	334,374
KZ	124,492	31,123
Mekkah	546,411	136,603

Table 1. Dataset sizes for each client

III) FEDERATED LEARNING METHODOLOGY

The global model is trained using the Federated Averaging (FedAvg) algorithm. In this approach, a central server coordinates training across multiple distributed clients.

During each communication round, the following steps occur:

1. The server distributes the current global model to all participating clients.
2. Each client trains the model locally using its private dataset.
3. The locally updated model parameters are sent back to the server.
4. The server aggregates the parameters using weighted averaging to update the global model.

The overall federated learning workflow used in this project is illustrated in Fig. 2.

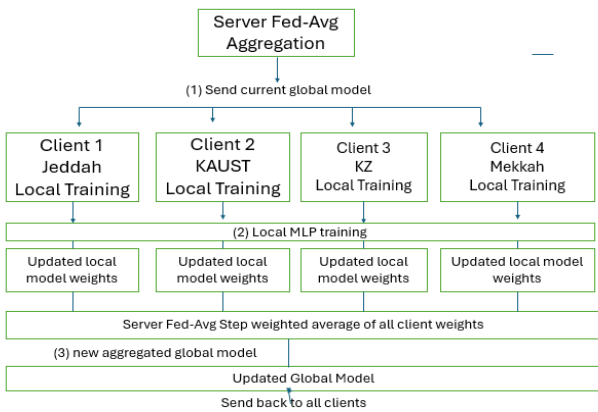


Fig. 1. Federated learning architecture used in this project. Each city acts as a federated client that performs local MLP training on private mobility data.

The updated model weights are sent to a central server where FedAvg aggregation is performed to update the global model.

The aggregation rule is defined as:

$$w_{(t+1)} = \sum_k (n_k / n) w_k$$

where w_k represents the model parameters from client k , n_k is the number of training samples at client k , and n is the total number of samples across all clients.

Deviation from the Source Paper

The first part of this work follows the Federated Averaging (FedAvg) framework presented in the reference paper. However, several practical adaptations were necessary due to the structure of the available dataset. First, the federated clients were defined at the city level rather than the user level. Second, because not all transportation modes were shared across all city datasets, the federated classification task was restricted to the common labels car and walk, resulting in a binary classification setting. These modifications were introduced to preserve label consistency across clients while maintaining the core federated learning procedure of local training followed by server-side FedAvg aggregation

IV) EXPERIMENTS

A. Experiment 1: FedAvg-Based Federated MLP Baseline

The first experiment implemented a baseline federated multilayer perceptron (FedMLP) model using the FedAvg algorithm. Each city acted as an independent federated client and participated in 10 communication rounds. The shared binary classification task was defined as car versus walk, since these were the only transportation modes common across all city datasets.

The final global performance of the baseline federated MLP model is summarized as follows:

- Global Accuracy: 0.6285
- Global Precision: 0.6733
- Global Recall: 0.8040
- Global F1-score: 0.7329

The final per-city results were:

- Jeddah: Accuracy = 0.3536, F1 = 0.4716
- KAUST: Accuracy = 0.7331, F1 = 0.8218
- KZ: Accuracy = 0.9046, F1 = 0.0020
- Mekkah: Accuracy = 0.5962, F1 = 0.7470

B- Experiment 2: Balanced Federated MLP

The second experiment investigated the effect of class imbalance on federated learning performance. Since the binary classes were not evenly distributed across all client datasets, the majority class in each client was downsampled to create a more balanced training set. The same federated MLP architecture and FedAvg aggregation procedure were then applied.

The final global performance of the balanced federated MLP model is summarized as follows:

- Global Accuracy: 0.5150
- Global Precision: 0.6776
- Global Recall: 0.4478
- Global F1-score: 0.5392

Compared with the baseline federated MLP, the balanced version achieved lower global accuracy and F1-score, suggesting that simple downsampling reduced the amount of useful training data and did not improve the overall federated model.

The final per-city results were:

- Jeddah: Accuracy = 0.5624, F1 = 0.4719
- KAUST: Accuracy = 0.5043, F1 = 0.6029
- KZ: Accuracy = 0.9046, F1 = 0.0020
- Mekkah: Accuracy = 0.4030, F1 = 0.4175

C. Experiment 3: Federated Ensemble Model

The third experiment extended the baseline federated framework by combining a federated MLP model with two locally trained models, namely Random Forest and Gradient Boosting. For each client, the final prediction was obtained using majority voting across the three models. The federated MLP component was still trained using the FedAvg algorithm, while the additional ensemble members were trained locally at the client side.

The final global performance of the federated ensemble model is summarized as follows:

- Global Accuracy: 0.9921
- Global Precision: 0.9961
- Global Recall: 0.9914
- Global F1-score: 0.9938

The final per-city ensemble results were:

- Jeddah: Accuracy = 1.0000, F1 = 1.0000
- KAUST: Accuracy = 0.9863, F1 = 0.9917
- KZ: Accuracy = 1.0000, F1 = 1.0000
- Mekkah: Accuracy = 0.9964, F1 = 0.9970

Model	Accuracy	Precision	Recall	F1-score
FedMLP baseline	0.6285	0.6733	0.8040	0.7329
Balanced FedMLP	0.5150	0.6776	0.4478	0.5392
Federated Ensemble	0.9921	0.9961	0.9914	0.9938

Table 2. Global Comparison Experiment Result

V) Discussion

The results of the first part demonstrate that federated learning can be successfully applied to distributed mobility datasets where each city acts as an independent client. The baseline FedMLP experiment achieved reasonable global performance, confirming that the FedAvg algorithm can train a shared binary classifier across geographically distributed datasets.

However, the results also showed substantial variation across clients. In particular, the KZ client achieved very high accuracy but extremely low F1-score in both the baseline and balanced federated MLP experiments. This indicates that accuracy alone is not sufficient to evaluate federated classification performance under strong class imbalance. The model likely favored the majority class in this client, resulting in misleadingly high accuracy.

The balanced federated MLP experiment did not outperform the baseline model. Although downsampling reduced class imbalance, it also removed a significant amount of training data, which appears to have weakened the global model.

The federated ensemble model achieved the strongest performance among all experiments. Nevertheless, the near-perfect results suggest that the current feature representation may allow the local models to strongly exploit location-based patterns. Since the available features were limited to Latitude, Longitude, and Time_seconds, the ensemble may be capturing spatial regularities rather than fully learning transportation behavior. This observation motivates the need for more advanced mobility-specific features in future work.

VI) Conclusion for Part 1

In the first part of this project, a federated learning framework for binary transportation mode classification was successfully implemented using city-level clients and the FedAvg algorithm. The shared classification task was defined as car versus walk, since these were the only transportation modes common across all participating city datasets. Three experiments were conducted: a baseline federated MLP model, a balanced federated MLP variant, and a federated ensemble model combining a federated neural network with local tree-based classifiers.

The results showed that the baseline FedMLP provided a valid federated learning starting point, while the balanced variant did not improve global performance. The ensemble model produced the strongest results, although further work is needed to verify whether these gains reflect genuine mobility behavior learning or strong dependence on limited spatial features. Overall, the first part of the project establishes a complete FedAvg-based binary federated learning pipeline that can serve as the foundation for the second part of the study. any figures or tables you use at the top or bottom of a column. Don't place them in the middle of a column. If particularly wide, a table or figure can span across both columns. Insert a table or figure after the point where it is first cited in the text.

Part 2

The first part of this project implemented a federated learning framework using the FedAvg algorithm for binary transportation mode classification across multiple city datasets. While this setup ensured label consistency across clients, it required restricting the classification task to only the transportation modes common across all cities, namely car and walk.

However, in real-world mobility scenarios, different cities often contain additional transportation modes that are not shared globally. For example, the KAUST dataset includes several additional modes such as bike, scooter, jog, motorcycle, and multiple bus types. These additional classes cannot be easily integrated into a single global federated model without introducing label inconsistency across clients.

Recent research in federated learning has addressed this limitation through personalized federated learning approaches, where a shared global model captures common patterns while individual clients maintain specialized local models for their unique data distributions (Dinh et al., 2020). Additionally, heterogeneous federated learning methods have been proposed to address differences in data distributions and label spaces across clients (Li et al., 2021).

Motivated by these approaches, the second part of this project investigates local and ensemble-based federated learning strategies for handling heterogeneous transportation mode datasets. Instead of restricting the classification task to shared labels, this part explores how city-specific models can be trained locally while still benefiting from federated-style collaboration.

The experiments in Part 2 focus specifically on the KAUST dataset, which contains a richer set of transportation modes and significantly larger sample sizes compared with other cities.

VII) KAUST DATASET AND MULTI-CLASS SETUP

Unlike the datasets used in Part 1, which were restricted to common labels across cities, the KAUST dataset contains eight transportation modes, enabling a more complex multi-class classification problem.

The transportation modes included in the KAUST dataset are:

- bike
- bus_colored
- bus_on demand
- car
- jog
- motorcycle
- scooter
- walk

After preprocessing and cleaning, the KAUST dataset contained:

Total samples: 3,923,657

Total features: 92

The class distribution is summarized in Table II.

Mode	Samples
walk	1,390,880
scooter	860,541
bike	592,532
jog	285,423
car	268,516
Bus on demand	236,655
Bus colored	160,737
motorcycle	128,373

Table 3. KAUST Transportation Mode Distribution

The dataset exhibits strong class imbalance, particularly for walking and scooter modes, which dominate the dataset. Because of this imbalance, evaluation metrics beyond simple accuracy are necessary. Following common practice in multi-class classification, macro-precision, macro-recall, and macro-F1 scores were used to better evaluate performance across all classes.

VIII) EXPERIMENT 4 – CENTRALIZED MULTI-CLASS BASELINE

To establish a performance baseline for the KAUST dataset, centralized models were trained using the full dataset. Two machine learning models were evaluated:

- Random Forest (RF)
- Multilayer Perceptron (MLP)

The Random Forest model is a tree-based ensemble method that combines predictions from many decision trees. Each tree is trained on a different random subset of the data and features, and the final prediction is obtained by majority voting across trees. Compared with neural networks, Random Forest is often better suited for structured tabular data because it can model non-linear feature interactions without requiring heavy feature scaling or extensive hyperparameter tuning. For this reason, it was used as the main classical baseline in Part 2.

The multilayer perceptron (MLP) is a feed-forward neural network composed of fully connected layers and non-linear activation functions. Unlike Random Forest, the MLP learns distributed feature representations through gradient-based optimization. It was included in this experiment as a neural baseline to compare tree-based learning with a standard deep learning approach. This comparison helps determine whether the KAUST feature space is better handled by neural networks or by classical ensemble models.

The dataset was divided into training and testing subsets using an 80/20 split. Due to the large dataset size, the MLP model was trained on a randomly sampled subset of the training data to reduce computational cost.

Random Forest Results:

The Random Forest classifier achieved strong performance across the multi-class dataset.

Results:

Accuracy: 0.9083
Macro Precision: 0.8953
Macro Recall: 0.8414
Macro F1-score: 0.8656

MLP Results:

The neural network model performed significantly worse than the tree-based model.

Accuracy: 0.5514
Macro Precision: 0.5281
Macro Recall: 0.2557
Macro F1-score: 0.2272

To better visualize the difference between the centralized models, Figure 5 presents the macro-F1 comparison between Random Forest, MLP, and the class-weighted Random Forest variant. Macro-F1 is used here because the KAUST dataset is strongly imbalanced across transportation modes, and this metric gives a more balanced view of performance across all classes.

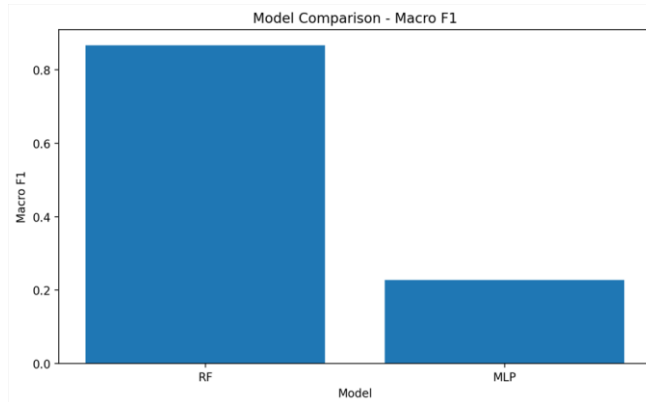


Fig. 2. Macro-F1 comparison of centralized KAUST models.

The figure clearly shows that the standard Random Forest achieved the strongest macro-F1 score, while the MLP performed substantially worse. The weighted Random Forest remained competitive but did not surpass the original Random Forest baseline. This confirms that, for the current KAUST tabular dataset, tree-based methods are more suitable than the neural baseline.

These results indicate that tree-based ensemble models are better suited for the current tabular mobility dataset, while the neural network struggled to learn effective representations from the available features.

Since the Random Forest achieved the best centralized performance, its confusion matrix is shown in Figure 6 to provide a more detailed view of class-wise behavior. Unlike scalar metrics such as accuracy, the confusion matrix allows inspection of which transportation modes are correctly separated and which modes are more frequently confused

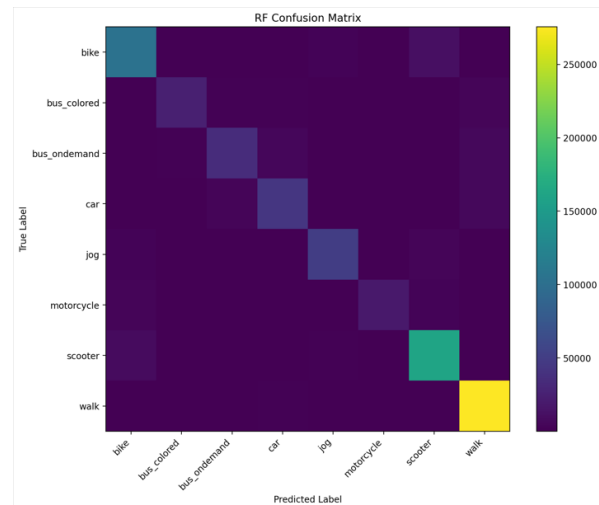


Fig. 3. Confusion matrix of the centralized Random Forest model on the KAUST dataset.

The confusion matrix shows that most correct predictions are concentrated along the diagonal, indicating strong overall classification performance. Off-diagonal values reveal the remaining classification errors and are particularly useful for understanding how minority or behaviorally similar classes are confused. This figure supports the use of the centralized Random Forest as the strongest baseline for the later distributed experiments.

Model	Accuracy	Macro Precision	Macro Recall	Macro F1
Random forest	0.9083	0.8953	0.8414	0.8656
MLP	0.5514	0.5281	0.2557	0.2272

Table 4. Centralized Model Comparison

X) EXPERIMENT 5 – CLIENT-LEVEL LOCAL MODELS

To simulate a federated learning environment, the KAUST dataset was partitioned into five independent client datasets. Each client represents a local data holder that trains its own model independently.

In this experiment, the same Random Forest architecture was used at the client level, but now each model was trained only on the local data available to one simulated client. Unlike the centralized baseline, each local model only sees a

subset of the full KAUST distribution. This setup makes it possible to measure how much performance is lost when the same model is trained in a decentralized environment instead of using all data centrally.

The dataset was split evenly across clients, producing approximately: ~784,000 samples per client

Each client then trained a local Random Forest classifier using its private data. Model evaluation was performed using an 80/20 train-test split within each client dataset.

The results for the local models are summarized in Table VI

Client	Accuracy	Macro Precision	Macro Recall	Macro F1
Client 1	0.8386	0.7977	0.7341	0.7611
Client 2	0.8378	0.7975	0.7318	0.7595
Client 3	0.8397	0.8025	0.7377	0.7650
Client 4	0.8391	0.8010	0.7347	0.7627
Client 5	0.8382	0.7968	0.7332	0.7603

Table 5. Client-Level Model Performance

Figure 7 visualizes the macro-F1 performance of the five local client models trained independently after splitting the KAUST dataset into equal-sized client partitions. This figure helps illustrate whether the clients behave similarly or whether significant variation appears across the local training settings.

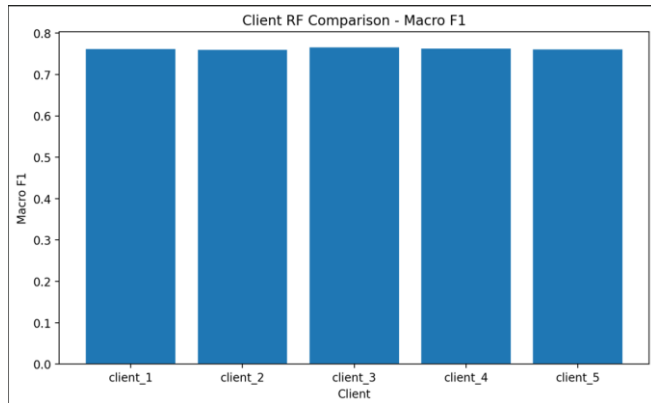


Fig. 4. Macro-F1 comparison across the five simulated local KAUST clients.

The figure shows that the local client models achieve relatively consistent macro-F1 scores, indicating that the equal random split produced client datasets with similar statistical properties. However, all client models remain below the centralized Random Forest baseline, which highlights the performance loss introduced by data decentralization

These results demonstrate consistent performance across all simulated clients, suggesting that the dataset partitions maintain similar statistical properties

XI) EXPERIMENT 6 – FEDERATED ENSEMBLE LEARNING

To approximate collaborative federated learning without directly sharing training data, an ensemble aggregation strategy was implemented.

The federated ensemble differs from both the centralized model and the isolated client models because it does not aggregate training data or model weights. Instead, each client first trains its own local Random Forest model, and the final prediction is obtained by majority voting across client predictions. This makes the method simpler than weight-based federated optimization while still allowing collaborative inference. The main purpose of this experiment is to test whether prediction-level aggregation can recover part of the performance lost due to decentralization.

Each client trained its own local Random Forest model. During inference, predictions from all client models were combined using majority voting, producing a federated ensemble prediction.

The ensemble model achieved the following results:

Accuracy: 0.8827
Macro Precision: 0.8719
Macro Recall: 0.7940
Macro F1-score: 0.8266

Metric	Value
Accuracy	0.8827
Macro Precision	0.8719
Macro Recall	0.7940
Macro F1	0.8266

Table 6. Federated Ensemble Performance

To compare isolated local learning with collaborative prediction, Figure 8 shows the macro-F1 scores of all client models alongside the majority-vote federated ensemble. This visualization is important because it shows whether combining local predictions can recover performance lost during client-level decentralization.

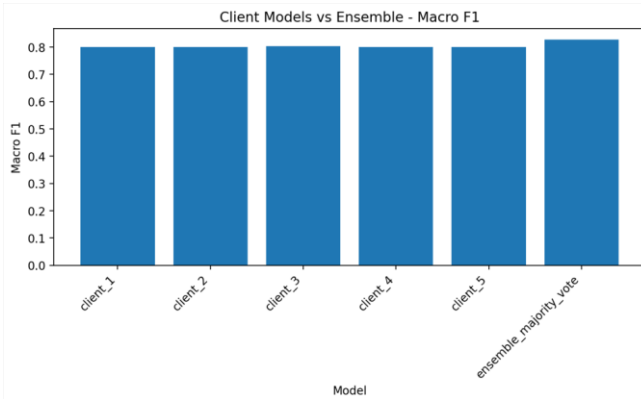


Fig. 5. Macro-F1 comparison between local client models and the federated ensemble (equal client split).

The figure confirms that the federated ensemble clearly outperforms all individual client models. Although it still remains below the centralized Random Forest baseline, the aggregated predictor recovers a substantial portion of the lost performance. This makes the federated ensemble a practical compromise between centralized accuracy and decentralized data locality.

Although slightly lower than the centralized Random Forest baseline, the ensemble model demonstrates strong performance while preserving data locality across clients.

XII) EXPERIMENT 7 – NON-IID CLIENT SIMULATION

Federated learning systems often operate under non-identically distributed (non-IID) data, where clients have different data distributions. This condition commonly occurs in mobility datasets because transportation patterns vary across geographic regions.

This experiment does not introduce a new model architecture; instead, it changes the data distribution across clients. In the earlier client experiment, the KAUST dataset was split approximately evenly, which produced relatively similar local datasets. In the non-IID setting, clients were intentionally assigned different dominant transportation modes and different dataset sizes. This makes the federated environment more realistic, since real-world clients rarely have identical class distributions.

To simulate this scenario, the KAUST dataset was partitioned into clients with uneven sample sizes, producing heterogeneous client datasets:

Client 1: 246,099 samples
 Client 2: 223,776 samples
 Client 3: 118,157 samples
 Client 4: 118,208 samples
 Client 5: 303,661 samples

Local models were trained for each client, and their predictions were aggregated using the same ensemble strategy.

The resulting ensemble performance was:

Accuracy: 0.8868
 Macro Precision: 0.8521
 Macro Recall: 0.7937
 Macro F1-score: 0.8196

Figure 9 presents the macro-F1 comparison for the non-IID experiment, where clients were intentionally constructed with skewed label distributions and uneven dataset sizes. This figure is used to evaluate whether the ensemble strategy remains effective when the local client distributions are more heterogeneous.

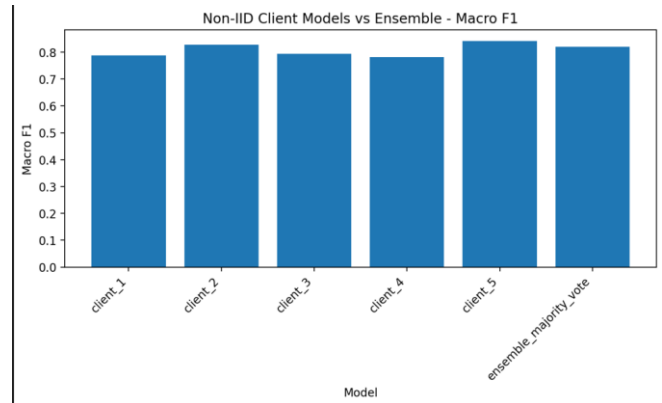


Fig. 6. Macro-F1 comparison between non-IID local client models and the non-IID federated ensemble.

The non-IID ensemble remains stronger than the isolated local client models, indicating that collaborative prediction is still beneficial under heterogeneous local data distributions. Although the macro-F1 score remains below the centralized baseline, the ensemble preserves reasonable stability even when the client datasets are no longer balanced or identically distributed.

The relatively stable performance under non-IID conditions suggests that ensemble-based federated strategies remain robust even when client datasets differ significantly in size and distribution.

XIII) EXPERIMENT 8 – CLASS-WEIGHTED RANDOM FOREST

To address the class imbalance observed in the KAUST dataset, a weighted Random Forest classifier was trained using class weights inversely proportional to class frequency.

The class-weighted Random Forest uses the same tree-based ensemble architecture as the original Random Forest, but it changes the training objective by assigning larger importance to underrepresented classes. In other words, classification errors on minority transportation modes contribute more strongly during tree construction. This

variant was introduced to test whether the imbalance visible in the KAUST dataset could be mitigated without changing the overall model family.

The weighted model achieved:

Accuracy: 0.9039
Macro Precision: 0.8933
Macro Recall: 0.8356
Macro F1-score: 0.8614

Method	Accuracy	Macro Precision	Macro Recall	Macro F1
Centralized Random Forest	0.9083	0.8953	0.8414	0.8656
Centralized MLP	0.5514	0.5281	0.2557	0.2272
Client-Level Local Models (Avg)	0.8389	0.7991	0.7343	0.7617
Federated Ensemble	0.8827	0.8719	0.7940	0.8266
Non-IID Federated Ensemble	0.8868	0.8521	0.7937	0.8196
Class-Weighted Random Forest	0.9039	0.8933	0.8356	0.8614

TABLE 7. PERFORMANCE COMPARISON OF ALL MODELS EVALUATED IN PART 2.

These results demonstrate that class weighting slightly improves fairness across minority classes while maintaining strong overall accuracy.

XIV) DISCUSSION

The experiments in Part 2 highlight several important observations regarding federated learning for heterogeneous mobility datasets.

The main difference between the Part 2 models lies in how much of the full KAUST data distribution they are allowed to observe and how collaboration is introduced. The centralized Random Forest is trained on the entire cleaned KAUST dataset and therefore provides the strongest upper-bound baseline. The local client models use the same learning algorithm, but each one only sees a fraction of the full data, which reduces performance. The federated ensemble does not retrain a single global model; instead, it combines predictions from multiple local models through majority voting, allowing partial performance recovery without centralizing raw data. Finally, the non-IID experiment keeps the same ensemble idea but makes the client distributions deliberately heterogeneous, which tests the robustness of collaborative prediction under more realistic federated conditions.

First, centralized Random Forest models achieved the strongest performance, indicating that tree-based ensemble methods are well suited for high-dimensional mobility datasets. However, centralized training violates the decentralized data ownership assumption that motivates federated learning systems. Neural network models performed substantially worse, likely due to the tabular structure of the data and the presence of strong feature correlations.

An additional observation from the experiments is the trade-off between centralized performance and federated data locality. While the centralized Random Forest achieved the highest accuracy, this setup requires aggregating all client data into a single location, which contradicts the privacy-preserving motivation of federated learning. In contrast, the federated ensemble approach demonstrated competitive performance while maintaining decentralized training across clients. This result suggests that collaborative prediction strategies can provide a practical compromise between model performance and privacy preservation in multi-city mobility datasets. Furthermore, the experiments confirm that heterogeneous transportation datasets, where different clients contain different label spaces, can still benefit from federated-style collaboration through local model aggregation. This finding aligns with recent personalized federated learning research, which emphasizes the importance of balancing shared knowledge with client-specific model specialization.

Second, local client models achieved relatively consistent performance across all simulated clients, suggesting that the KAUST dataset maintains stable statistical properties when partitioned.

Third, the federated ensemble strategy provided strong performance while preserving client data locality. This approach aligns with recent work in personalized federated learning, where local models capture client-specific characteristics while still contributing to a global system (Dinh et al., 2020).

Finally, experiments with non-IID data confirmed that ensemble-based federated approaches remain robust under heterogeneous data distributions, which is a common challenge in real-world federated learning systems (Li et al., 2021).

XV) CONCLUSION FOR PART 2

The second part of this study investigated strategies for handling heterogeneous transportation mode datasets that contain city-specific labels. Using the KAUST dataset, multiple experiments were conducted including centralized training, client-level local models, federated ensemble aggregation, non-IID client simulation, and class-weighted training.

The results show that local and ensemble-based approaches can effectively handle multi-class mobility classification

while maintaining the distributed nature of federated learning systems.

Together with the FedAvg framework developed in Part 1, these experiments demonstrate how federated learning systems can be extended to support heterogeneous transportation datasets where clients may contain different label spaces and mobility patterns.

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